

Experiment-2

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Course Code: -MLA0305

Slot-B

Title: - Reinforcement Learning framework using a Markov Decision Process (MDP)

AIM: - To implement a Reinforcement Learning framework using a Markov Decision Process (MDP) and demonstrate the interaction between an agent and environment through states, actions, transitions, and rewards using Q-Learning.

PROCEDURE

1. Import the required Python libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn.
2. Generate a synthetic dataset using the Gaussian Mixture Model (GMM).
3. Generate 500 records containing:
4. Display the first 10 samples of the generated dataset.
5. Calculate and display the dataset summary containing minimum, maximum, mean, and standard deviation.
6. Save the generated dataset as a CSV file and provide an option to download it.
7. Define the MDP states from 0 to 9.
8. Define four possible actions:
9. Define the transition function to determine the next state and reward for each state-action pair.
10. Initialize the Q-table with zeros.
11. Set the Q-Learning parameters:
 - Learning rate, $\alpha = 0.10$
 - Discount factor, $\gamma = 0.90$
 - Exploration rate, $\epsilon = 0.20$
12. Train the Q-Learning agent for 500 episodes.
13. Use the ϵ -greedy policy to balance exploration and exploitation.
14. Update the Q-values using the Q-Learning Bellman equation:
$$Q(s,a)=Q(s,a)+\alpha[r+\gamma\max_{a'}Q(s',a')-Q(s,a)]$$
15. Generate and display the learned Q-table.
16. Determine the optimal action/policy for each state.
17. Run the trained agent through the MDP environment and record its state path and rewards.
18. Create a random agent and evaluate its performance.
19. Compare the random agent and trained Q-Learning agent using average, maximum, and minimum rewards.
20. Generate visualizations for: GMM reward distribution, Q-Learning training performance, Steps taken during training, Q-table heatmap, Random agent vs trained agent comparison
21. Display the final summary table and performance results.

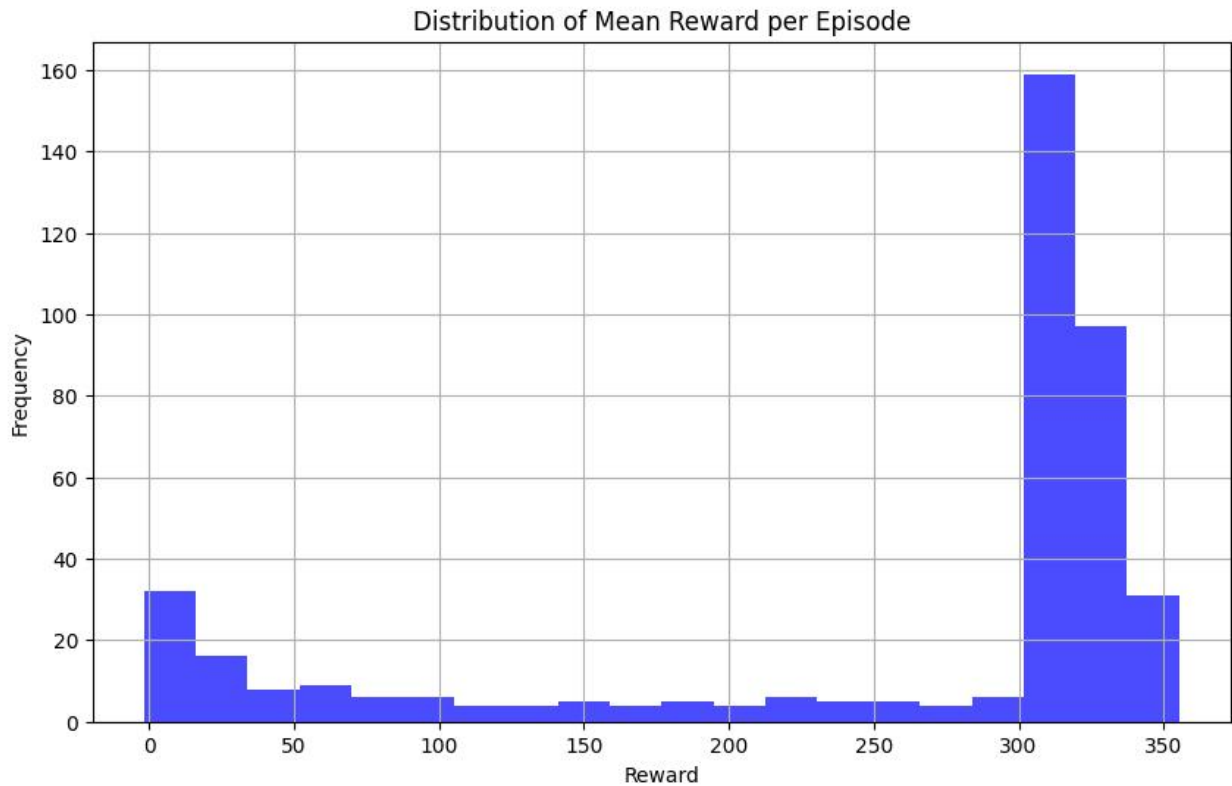


FIG: -1 MEAN REWARD PER EPISODE

State	Action	Reward	Return
3	Right	-2	5
4	Left	-1	7
3	Right	-2	8
4	Right	10	10

State	Action	Reward	Return
3	Right	-2	5
4	Up	-1	7
1	Down	-2	8
4	Right	10	10

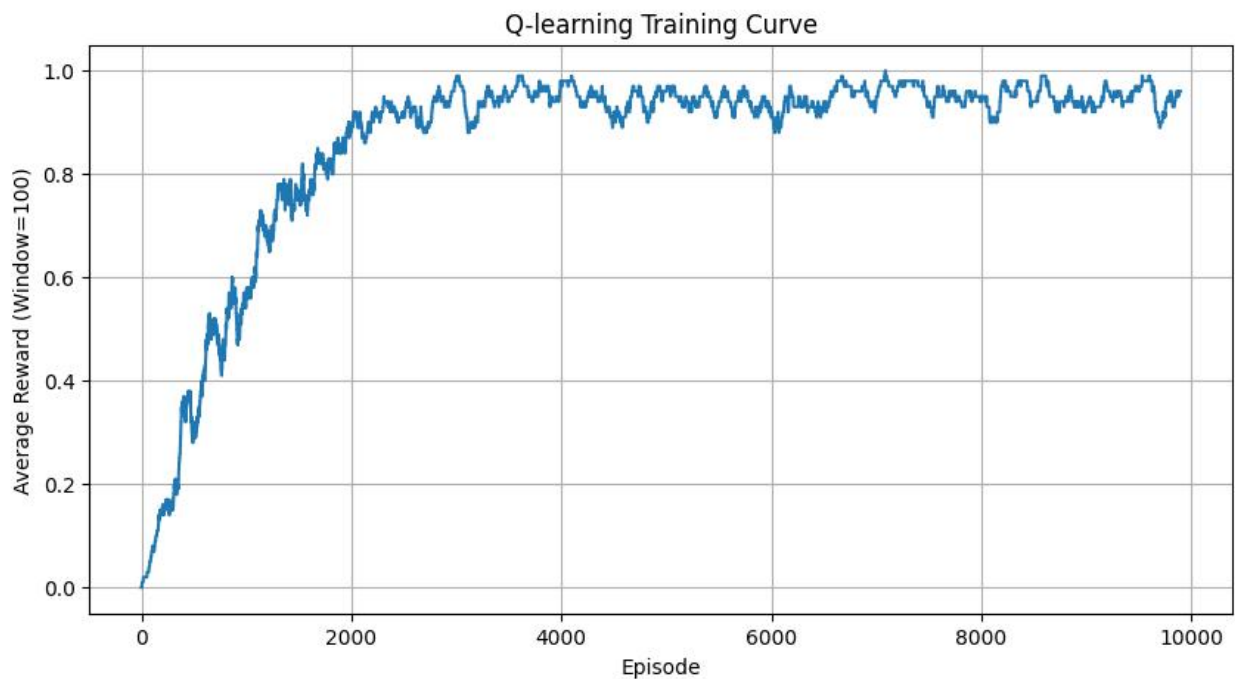
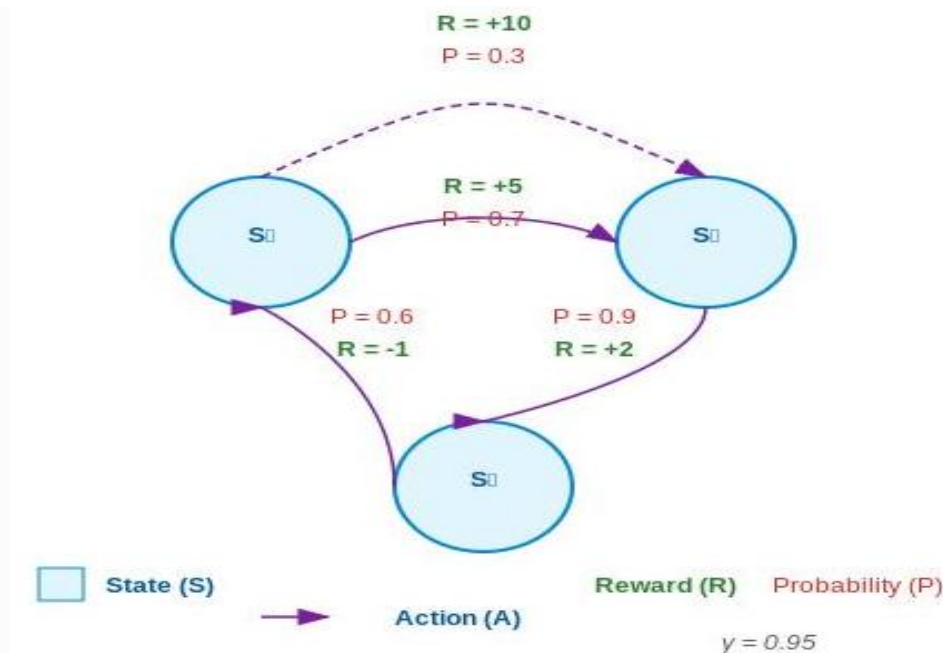


FIG: -4 Q LEARNING CURVE

RESULT

The Reinforcement Learning framework and Markov Decision Process were successfully implemented using Q-Learning.